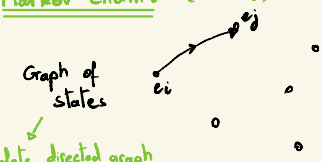
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1905 – Andres Markov (the grandfather of generative AI)

Markov chain: $P(e_i \to e_j)$



Complete directed graph

$V(e_i, e_j)$: prob(e_i, e_j)

Hypotheses: Not constantly moving

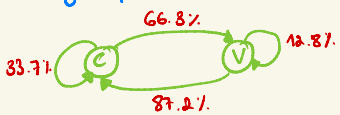
1. All states are known (Universe is known, stable, closed)

example:

- Assume a Universe U a set of sequences created by:
  - C – Consonant
  - V – vowel

1) → Approximate U with cc, cv, vc, vv in Pushkin's poetry

2) → Learning step



C	V	cc	cv	vc	vv
56.8%	43.2%	33.7	66.3	87.2	12.8

3) → Generative step

- Generate $S = \{ccevc\ vccnvc \ldots.\}$ Similar to pushkin's

should be similar to this

S_c	S_v	S_{cc}	S_{cv}	S_{vc}	S_{vv}
56.8%	43.2%	33.7	66.3	87.2	12.8

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SHANNON : Father of AI

- Assume a Universe U a set of sequences created by BIGRAMS.

$B_{26 \times 26} = \{aa, ab, ac, \dots\}$

$G = (V, E)$ with $V = 26$ and $E = 26^2 \times 26^2$

→ Approximate U with $3.5 \cdot 10^{12}$ letters (learning data)

→ P. Norwig learned the frequence of diffrent BIGRAMS

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